6.2.4 Containerizing Spring Boot with Cloud Native Buildpacks

Cloud Native Buildpacks (https://buildpacks.io) is a project hosted by the CNCF to "transform your application source code into images that can run on any cloud." When introducing containers in chapter 1, I underlined how PaaS platforms like Heroku and Cloud Foundry are actually using containers behind the scenes, converting your application source code into containers before running them. Buildpacks is the tool they use to accomplish that.

Cloud Native Buildpacks has been developed and advanced based on Heroku and Pivotal's many years of experience running cloud native applications as containers on their PaaS platforms. It's a mature project, and since Spring Boot 2.3, it has been integrated natively in the Spring Boot Plugin for both Gradle and Maven, so you're not required to install the dedicated Buildpacks CLI (`pack`).

These are some of its features:

- It auto-detects the type of application and packages it without requiring a Dockerfile.
- It supports multiple languages and platforms.
- It's highly performant through caching and layering.
- It guarantees reproducible builds.
- It relies on best practices in terms of security.
- It produces production-grade images.
- It supports building native images using GraalVM.

NOTE If you'd like to know more about Cloud Native Buildpacks, I recommend watching "Cloud Native Buildpacks with Emily Casey" (http://mng.bz/M0xB). Emily Casey is a member of the Buildpacks core team.